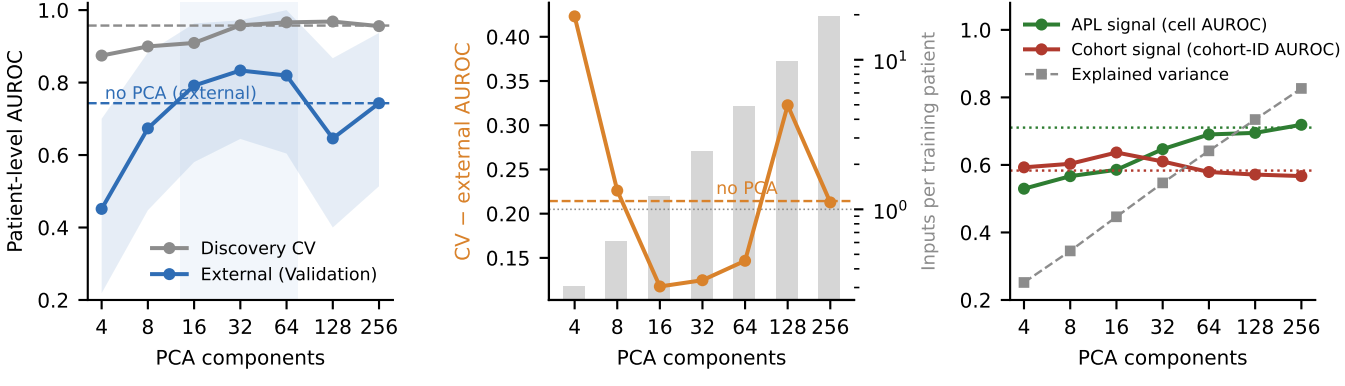
(a) Patient-level APL detection
Patient-level AUROC
1.0
0.8
0.6
0.4
0.2
no PCA (external)
Discovery CV
External (Validation)
4
8
16
32
64
128
256
PCA components
(b) Generalization gap and model size
CV − external AUROC
0.40
0.35
0.30
0.25
0.20
0.15
no PCA
$10^1$
$10^0$
Inputs per training patient
4
8
16
32
64
128
256
PCA components
(c) What the components keep
APL signal (cell AUROC)
Cohort signal (cohort-ID AUROC)
Explained variance
1.0
0.8
0.6
0.4
0.2
4
8
16
32
64
128
256
PCA components